STREAMPROOF: A TRUST AND PROVENANCE LAYER FOR CITIZEN STREAM REPORTS ON HL7 FHIR

A MINI PROJECT REPORT

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ABSTRACT

Citizens routinely photograph foamy, stagnant or foul-smelling urban streams, but a city cannot tell which of these reports to trust, a public-health system cannot read them, and nothing states what a report is *allowed to be used for*. To a computer, a rumour and an expert-verified observation look exactly the same. This mini project presents **StreamProof**, a trust and provenance add-on to the HL7 Europe OneAquaHealth (OAH) FHIR Implementation Guide, built for Track 7 (Digital Health Standards) of the OneAquaHealth IEEE Global Hackathon.

StreamProof grades every citizen report from A to D using seven transparent, rule-based checks (photo quality, photo time, GPS accuracy, nearness to a mapped stream, nearby agreement, real rainfall context and observer track record) and attaches a plain-language reason to every point. Reports then climb an evidence graph: Report, Assessed, Community-supported, Expert-verified and Decision-grade. A **permitted-use gate** decides, for each output, what a record at each level may be used for. The gate is not hidden in code: it is loaded from the published FHIR CodeSystem `trust-level`, so any receiving system can enforce the same rule. Verified records leave as FHIR R4 Bundles whose profiles derive from the official OAH profiles and carry grade, trust level and permitted uses inside the record, with an Ed25519-signed, de-identified certificate.

The system is implemented as a Python/FastAPI backend, a Next.js progressive web app in 45 languages, and FSH-defined FHIR profiles. The repository's recorded HL7 Validator run shows the FHIR output conforming to the official OAH profiles with 0 errors and 0 warnings and an unverified record rejected by the profile itself, and in our own run the backend test suite of 93 tests gave 88 passing and 5 skipped. The accuracy of the grade against real expert decisions has **not** been measured because no real field data existed; this limitation is stated openly in the report.

Keywords: Citizen science, trust, provenance, HL7 FHIR, OneAquaHealth, permitted-use gate, evidence grading, urban streams, interoperability.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CSV	Comma-Separated Values
DSS	Decision Support System
EXIF	Exchangeable Image File Format
FHIR	Fast Healthcare Interoperability Resources
FSH	FHIR Shorthand
GDPR	General Data Protection Regulation
GPS	Global Positioning System
HL7	Health Level Seven International
IG	Implementation Guide
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
OAH	OneAquaHealth
OSM	OpenStreetMap
PDF	Portable Document Format
PWA	Progressive Web App
REST	Representational State Transfer
SHA	Secure Hash Algorithm
SQL	Structured Query Language
UCUM	Unified Code for Units of Measure
UI / UX	User Interface / User Experience
WCAG	Web Content Accessibility Guidelines

CHAPTER 1
INTRODUCTION

1.1 Background

Urban streams are the most visible part of a city's water system, and the people who live beside them are often the first to notice that something has changed: foam that does not break up, green scum, a sewage smell, ponded water full of mosquitoes, or dead fish. Citizen-science platforms already let residents photograph and report such signs. The difficulty begins after the report is made. A municipality that receives hundreds of photographs must decide which ones deserve a field visit, a public-health team needs records in a format its systems can read, and an ecologist needs to know how much weight a single observation can carry.

OneAquaHealth (OAH) is a European Union funded project (grant 101086521) that links the health of urban aquatic ecosystems with human health. Its outputs include a Catalogue of Measures for rehabilitating urban water bodies, a mosquito (Diptera) prediction model called DipteraCAST, a Citizen Science App and an HL7 FHIR Implementation Guide (IG) maintained by HL7 Europe. FHIR (Fast Healthcare Interoperability Resources) is the standard that health information systems use to exchange structured records, and it is therefore the natural language in which stream observations should reach a public-health partner.

StreamProof sits in the gap between a citizen's phone photograph and a FHIR record that a health system can safely act on. It does not replace the OAH guide; it adds a *trust and provenance* layer on top of it.

1.2 Problem statement

A citizen report today carries no machine-readable answer to three questions that every downstream user needs answered:

- **How much should this report be trusted?** A blurry photo taken five days ago, a report with no GPS accuracy and a report corroborated by three different neighbours are stored in the same way, so a city cannot rank them.
- **Who or what vouches for it?** Provenance (who observed, which engine assessed it, which expert verified it, under which rule) is normally lost when a report is exported.
- **What may it be used for?** A rumour may justify sending someone to look; it should not trigger a public-health advisory. Nothing in the current exchange formats says which use is acceptable at which level of evidence.

The problem addressed by this project is therefore to design and build a system that grades citizen stream reports with explainable reasons, lets neighbours and experts strengthen them, and writes the resulting trust information *inside* the standard FHIR record together with a published, enforceable rule for permitted use.

1.3 Motivation and Significance

Evidence that cannot be trusted is either ignored, which wastes the effort of the citizens who produced it, or over-trusted, which risks false alarms. Both outcomes discourage participation. The literature on citizen-science data quality, notably Alabri and Hunter [1] and the review by Baker *et al.* [2], shows that automated checks, community agreement and expert verification each have a role, and recommends a hierarchy in which most records are checked automatically or by consensus and experts examine the flagged ones. What those works do not provide is a way to carry the outcome of that hierarchy across system boundaries and to restrict how the record may be used afterwards.

The significance of StreamProof lies in making two things explicit that are normally implicit: the *trust level* of a record, and the *permitted uses* at that level. Because both are published as FHIR CodeSystems, a receiving system does not have to take the sender's word for it; it can enforce the same rule independently.

1.4 Project Overview

StreamProof is a full-stack prototype consisting of a Python backend (evidence engine, evidence graph, permitted-use gate, FHIR packaging, signing and reporting), a Next.js progressive web app for citizens and for organisation reviewers, and a set of FHIR Shorthand (FSH) definitions that compile into profiles derived from the official OAH profiles. It was built for Track 7, Digital Health Standards, of the OneAquaHealth IEEE Global Hackathon [12], and its source is published under the MIT licence [13].

The prototype runs on real data wherever real data exists: stream geometry for five European cities comes from OpenStreetMap, rainfall comes from Open-Meteo, and the recommended next steps in the River Health Brief are taken from the real OneAquaHealth Catalogue of Measures [3]. The demonstration reports and demonstration observers are synthetic and are labelled as such.

1.5 System Functionality

The system serves four groups of users, and each has its own functions.

- **Citizens** report in about a minute: take a photograph inside the page, tap what they see from ten plain-language signs (or "Everything looks fine"), and confirm the location, which snaps

to the nearest mapped stream. The report is graded immediately, with a reason for every point, and the citizen can follow its progress, receive a signed certificate once it is verified, and erase their identity at any time.

- **Reviewers (experts)** work from a prioritised queue. For each report they see the grade, the seven reasons, the nearby reports, a map and the permitted-use gate. They can verify (by remote or field check), reject with a reason the citizen can read, or ask neighbours for more evidence through a community mission placed about 400 m upstream.
- **Cities and public-health partners** receive the River Health Brief: signals grouped by sign and place, how sure the system is, recommended next steps quoted from the OAH Catalogue of Measures with section and page, advisory flags only at decision grade, and a list of stretches that nobody has observed recently.
- **Standards and data consumers** can export an expert-verified record as a FHIR R4 Bundle, export verified mosquito observations as ground truth for DipteraCAST (CSV or FHIR), read the published rule files, and verify a signed certificate (any edit to a value is detected).

1.6 Objectives

- To grade each citizen stream report from A to D with rule-based, human-readable reasons, so that every point can be argued with.
- To model trust as an evidence graph (Report, Assessed, Community-supported, Expert-verified, Decision-grade) in which several independent people, or one expert, can strengthen a report.
- To publish, as machine-readable FHIR data, what each trust level may be used for, and to make the running system enforce that same file.
- To package verified reports as FHIR R4 Bundles that conform to the official OAH profiles, and to make the profile itself reject unverified records.
- To protect citizens: pseudonyms only, positions coarsened to about 100 m, photograph metadata removed, and erasure that does not break the signed evidence.
- To help cities see where evidence is thin so that observation does not only follow busy, well-walked paths.
- To be honest about what has and has not been measured.

1.7 Design Philosophy and Approach

Four principles shaped the design. First, **explainability over accuracy claims**: the grade is rule-based on purpose, so that a citizen or an ecologist can read the sentence behind each point; an optional AI photograph suggestion is shown to the expert but never changes the grade. Second, **the rule is data, not code**: the permitted-use matrix is the published CodeSystem `trust-level` and the gate loads that file at start-up, so the published rule and the running rule cannot disagree. Third, **trust travels with the record**: grade, trust level and permitted uses are components of the FHIR Observation itself. Fourth, **privacy by design**: no names, contact details or exact coordinates ever appear in an exported record.

The approach was also shaped by a deliberate comparison with existing work. Research-grade consensus on a platform such as iNaturalist tells you that several people agree, but not what the record may be used for. Alabri and Hunter [1] combine automatic checks with a trust measure for each observer but do not publish an enforceable use rule or carry trust inside a health-data standard. Baker *et al.* [2] describe a verification hierarchy that StreamProof implements and extends with the use rule. No claim is made that StreamProof out-performs any of these at species identification, and no such benchmark was run.

1.8 Broader Impact

For a municipality, the immediate benefit is a ranked, explained list of what to check first and, once enough evidence exists, a decision-grade signal with next steps drawn from a recognised catalogue. For public-health partners, the benefit is that records arrive in FHIR with their trust and permitted use attached, so a mosquito observation can be used as DipteraCAST ground truth only when it deserves to be. For citizens, the benefit is feedback: their report’s status moves, an expert explains a decision, and a mission appears near them, with no points for volume and no ranking between people. The coverage view of the brief addresses fairness of attention along the stream, although it cannot address fairness between neighbourhoods because StreamProof holds no demographic data.

1.9 Educational Value

The project brings together several areas of the Information Technology curriculum in one working system: REST API design, relational storage, image analysis, geospatial computation (snapping to polylines and walking upstream), applied cryptography (SHA-256 hashing and Ed25519 signatures), health-data standards (FHIR profiling with FSH, ConceptMaps and CodeSystems), progressive web app engineering with offline support and internationalisation, automated testing, and privacy engineering. It also demonstrates how to evaluate honestly: the repository reports what

the rules do, what they cannot show, and what would have to be measured before an accuracy figure could be quoted.

1.10 Conclusion

This chapter introduced the problem of untrusted and unusable citizen stream reports, the OneAquaHealth setting, and the StreamProof approach of grading, corroborating, verifying and gating reports while carrying trust inside a standard FHIR record. The next chapter describes the tools and technologies chosen to build it.

CHAPTER 2
TOOLS AND TECHNOLOGIES USED

2.1 Python 3 and FastAPI

The backend is written in Python 3.11 or later. **FastAPI** exposes the REST API under `/api` and is served with Uvicorn; it was chosen for its typed request handling, its simple testing through a test client and its small footprint. Persistence uses **SQLite** through the Python standard library, which keeps the prototype installable with two commands. The numerical and imaging work uses **NumPy** and **Pillow**: Pillow reads the photograph, extracts the EXIF capture time and removes metadata from the stored copy, and NumPy computes sharpness (variance of a Laplacian filter) and brightness. Signing uses the **cryptography** library (Ed25519), the downloadable certificate is drawn with **ReportLab**, sessions use signed cookies (Starlette session middleware with `itsdangerous`), and reviewer passwords are hashed with scrypt and a per-user salt. The backend test suite uses **pytest** and **httpx**.

2.2 Next.js, React, TypeScript and Leaflet

The user interface is a Next.js (version 16) and React (version 19) application written in TypeScript. It is an installable **progressive web app**: a service worker caches the application shell, serves the last copy of pages and API reads when the network is unavailable, and never caches writes, while an outbox in the app stores reports made offline until the connection returns. Maps use **Leaflet** with OpenStreetMap tiles. The interface is translated into 45 languages, including right-to-left scripts, using locale files in `web/ui/locales`. Browser-level checks (end-to-end flow on a phone viewport, axe-core accessibility audit in light and dark mode, offline and service-worker check) are scripted with **Playwright**.

2.3 HL7 FHIR R4, FSH and the HL7 Validator

Interoperability rests on **HL7 FHIR R4 (4.0.1)** [5] and on the HL7 Europe OneAquaHealth guide [4]. The StreamProof add-on is written in **FHIR Shorthand (FSH)** [6] in `ig/input/fsh`, which defines the profiles `StreamProofObservation` (derived from `ObservationIndicatorsOah`), `StreamProofLocation` (derived from `LocationOah`) and `StreamProofProvenance`, the CodeSystems `trust-level`, `permitted-use`, `evidence-grade`, `evidence-attribute`, `citizen-sign` and `proposed-oah-indicator`, and a ConceptMap from citizen signs to OAH indicators. A build script compiles them with the OAH guide (fetched at build time rather than copied, because that guide carries no licence file), and the **HL7 Validator** (version 6.10.4 in the

recorded run, which needs Java) checks the produced Bundles against the official OAH profiles. Source code is edited in **Visual Studio Code** and versioned with **Git** and GitHub.

2.4 Summary

Table 2.1: Technology stack of StreamProof

Layer	Tool / library	Used for
Backend	Python 3.11+, FastAPI, Uvicorn	REST API, business rules
Storage	SQLite (standard library)	Reports, accounts, identity vault
Imaging	Pillow, NumPy	EXIF time, sharpness, brightness, metadata stripping
Security	cryptography (Ed25519), scrypt, itsdangerous	Signed certificates, password hashing, sessions
Documents	ReportLab	Signed contribution certificate (PDF)
Frontend	Next.js 16, React 19, TypeScript, Leaflet	Citizen PWA and organisation portal
Offline	Service worker, outbox	Offline reading and queued reports
Standards	HL7 FHIR R4, OAH IG, FSH, HL7 Validator	Profiles, CodeSystems, validation
Data	OpenStreetMap, Open-Meteo, OAH Catalogue	Streams, rainfall, recommended measures
Testing	pytest, httpx, Playwright, axe-core	Backend and browser checks

Every tool above is open source or freely available, which matches the project’s aim of publishing the rules as open data that outlive any single application.

CHAPTER 3
SYSTEM IMPLEMENTATION

3.1 Dataset

StreamProof is rule-based and does not train a machine-learning model, so there is no training dataset in the usual sense. The data it uses fall into five groups, summarised in Table 3.1.

Table 3.1: Data used by StreamProof

Data	Source	Content in this prototype
Stream geometry	OpenStreetMap (ODbL)	Five streams: Ribeira de Coselhas (Coimbra, 164 vertices), Sabato (Benevento, 109), Leie (Ghent, 131), Akerselva (Oslo, 153) and Hers-Mort (Toulouse, 173); more cities can be registered by search
Rainfall	Open-Meteo (CC BY 4.0)	Daily rainfall files for the five cities, used for the weather-context check
Measures	OneAquaHealth Catalogue of Measures, D2.4 (CC BY 4.0)	Twelve measures with section, title, page, what they do and their stated limits, matched to signs
FHIR definitions	Written in FSH, compiled	Fourteen definitions and five example Bundles, including one negative control
Demonstration reports	Synthetic, labelled as such	Fifteen seeded reports on the real Ribeira de Coselhas: 11 expert-verified, 2 community-supported, 1 not confirmed, 1 assessed (grades 4 A and 11 B)

The citizen vocabulary has ten signs: scum or green water, bad smell, dead fish, oily sheen, stagnant water, many mosquitoes, sewage or discharge, litter, foam and “Everything looks fine”. Six of them map to an indicator that already exists in the OAH code system (foam, fish, diptera, hydrology); four have no OAH concept yet (oil sheen, sewage, litter and the general visual check) and use proposed codes that the add-on offers back to HL7 Europe. The ConceptMap records the mapping as *wider*, because the OAH indicator is the broader concept; the indicator becomes the Observation code and the citizen’s sign becomes its value.

Note on the data. The seeded observers, reports and verifications are invented for demonstration. No real field photographs and no expert-labelled field data were available, which is why Chapter 4 does not quote an accuracy figure.

3.2 System Architecture

Figure 3.1 shows the layered architecture. Two web clients (the citizen PWA and the organisation portal) talk to a FastAPI layer that handles sessions, roles, rate limits, consent and photograph metadata stripping. Behind it, the evidence engine grades a report, the evidence graph moves it between trust levels, and the permitted-use gate sits in front of every output.

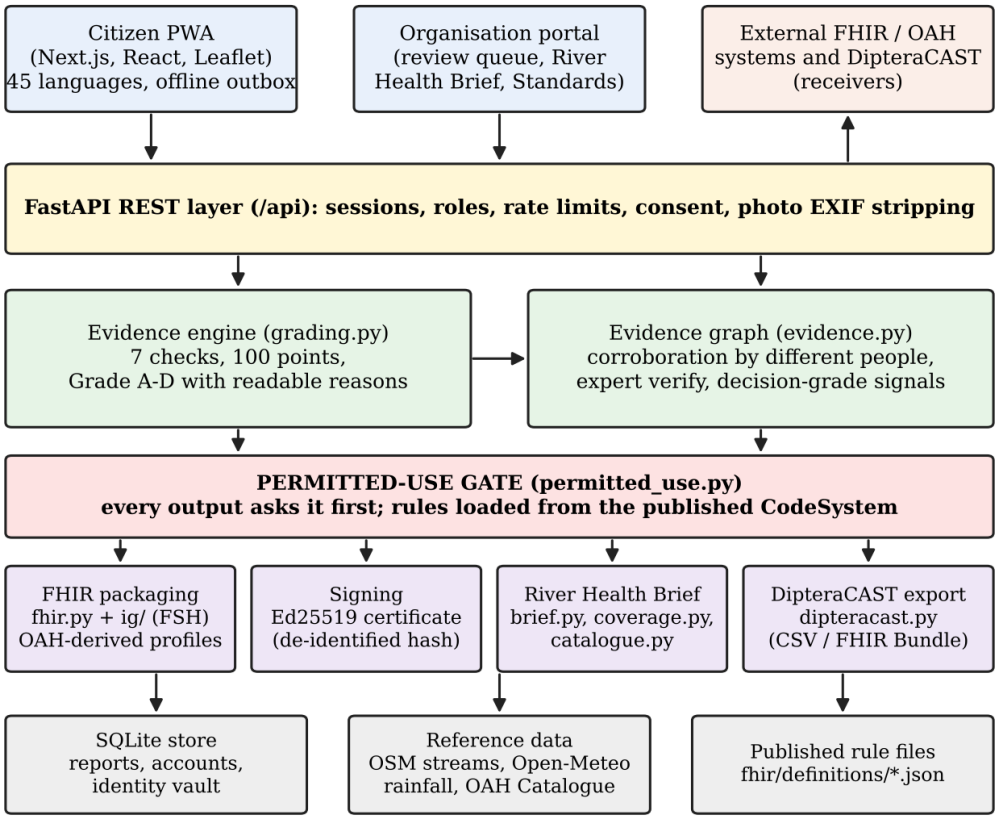


Figure 3.1: Layered architecture of StreamProof

Trust levels form a graph rather than a strict ladder (Figure 3.2). A report becomes *Assessed* when graded, *Community-supported* when independent people agree, *Expert-verified* when an expert confirms it and *Decision-grade* when a cluster of reports reaches the threshold. Expert verification is reachable directly from Assessed, so a rural report with no second observer is not stuck. A rejected report becomes *Not confirmed*, with a reason the citizen can read, and is kept for the record only.

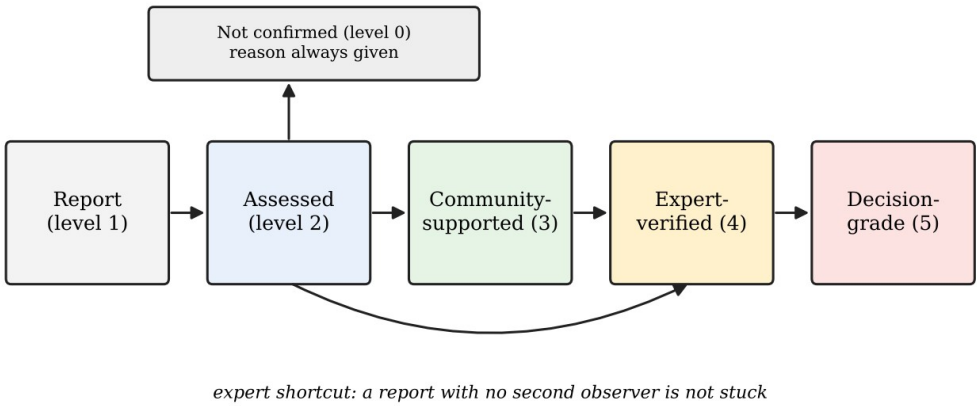


Figure 3.2: Trust levels and what each level adds to the permitted uses

Figure 3.3 follows one report from the phone to a signed FHIR Bundle.

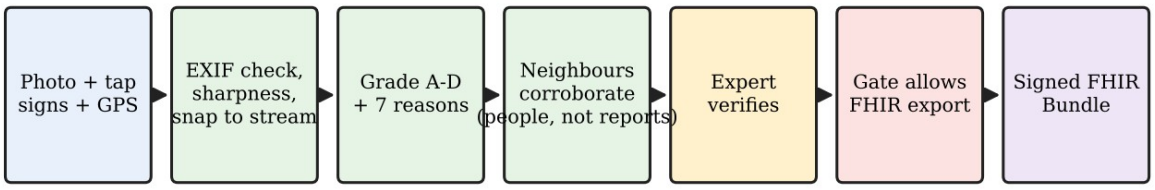


Figure 3.3: End-to-end path of one report

Table 3.2 is the permitted-use matrix. It is cumulative: each level keeps what the previous level allowed and adds the uses listed.

Table 3.2: Permitted-use matrix (published as the CodeSystem trust-level)

Trust level	Level	What it adds
Report	1	Triage queue, field-check request, community mission
Assessed (grade A–D with reasons)	2	Organisation dashboard, labelled unverified
Community-supported	3	Public map, shown as “reported, being checked”
Expert-verified	4	OAH dashboards, FHIR exchange, DipteraCAST ground truth, contributor recognition
Decision-grade	5	Advisory flag to agencies and public-health partners
Not confirmed	0	Nothing; kept for the record with a reason

3.3 Modules Description

3.3.1 Citizen reporting and photograph analysis

A report consists of one or more signs, an optional description, a location and an optional photograph. The photograph is analysed in `imaging.py`: its shortest side must reach 480 pixels, its sharpness (variance of the Laplacian) must reach 60, and its brightness must lie between 35 and 225; its EXIF capture time is read for the time check, and all EXIF data, including GPS, is removed from the stored copy. The report location is snapped to the nearest mapped stream by `geo.py`, which also provides the distance function, the 100 m coarsening used on every public surface and the upstream walk used by missions. Posting a report requires recorded consent and is rate limited per account.

3.3.2 Evidence engine (grade A–D)

`grading.py` scores a report out of 100 points using seven checks, each of which returns a status, a number of points and a sentence in plain words. The bands are A from 85, B from 70, C from 50 and

D below 50. A report with no photograph is capped at C, because a photograph is the most informative single item. Figure 3.4 and Table 3.3 show the weights.

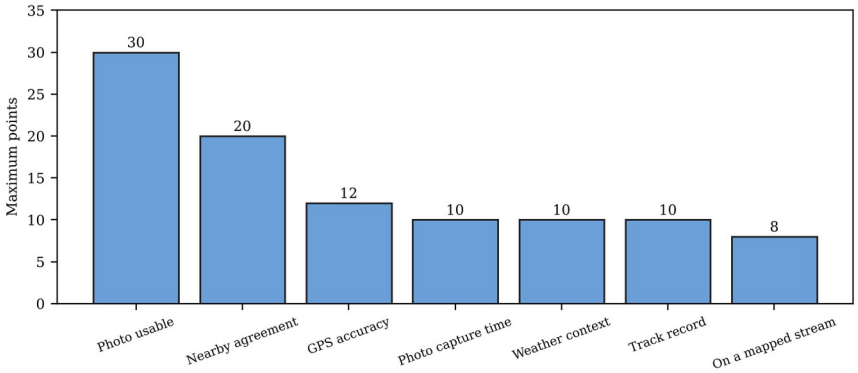


Figure 3.4: Maximum points of the seven grading checks

Table 3.3: The seven grading checks

Check	Max	Example reason shown to the citizen
Photo usable (size, sharpness, exposure)	30	Photo is clear and well exposed.
Nearby agreement or contradiction	20	1 other person reported the same within 500 m.
GPS accuracy	12	Location confirmed (GPS accuracy 7 m).
Photo capture time vs report time	10	Photo has no capture time (common after messaging apps).
Weather context (real rainfall)	10	Consistent with weather: 4.5 mm of rain in 7 days.
Observer track record	10	New observer: no history yet.
On a mapped stream	8	On Ribeira de Coselhas (11 m from the mapped channel).

The weather check is deliberately modest. Signs that favour dry conditions (for example stagnant water) score fully after a dry week and lose points after 10 mm or more of rain in 48 hours, because rain flushes ponded water; signs that rain tends to bring (for example sewage overflow) score fully after 5 mm or more. When no weather data exists for a place, the check scores 5 of 10 and says so.

3.3.3 Evidence graph and corroboration

`evidence.py` contains pure functions over lists of reports so that the rules can be tested without a database. Two reports corroborate each other only if they were made by *different people*, show the same sign within 500 m and 14 days, are graded C or better and are independent: at least 30 minutes apart in time or 50 m apart in place, which stops a group standing in one spot and tapping submit. Each account’s corroborations are capped at three per day, and each link is weighted by the corroborator’s track record (1.5 for a proven observer, 1.0 for a new one, 0.5 for one whose reports were mostly not confirmed). A total support of 1.0 lifts a report to Community-supported.

A *signal* is a cluster of reports of the same sign. A signal becomes **decision-grade** when it contains at least two expert-verified observers, or at least three observers at community-supported level or better, together with at least one expert; verified reports in such a cluster are promoted to Decision-grade, and a health-relevant sign then raises an advisory flag. Counting is by distinct observers, not by reports, so one person reporting twice is one voice.

3.3.4 Permitted-use gate

`permitted_use.py` loads the compiled CodeSystem files `CodeSystem-trust-level.json` and `CodeSystem-permitted-use.json` when it is imported. It builds the matrix from the `permits` and `level` properties, fails loudly if the file names an unknown use or omits a trust level, and exposes `check()` and `require()`. Every output (dashboard, public map, FHIR export, certificate, advisory flag, DipteraCAST export) asks the gate first and receives either permission or a reason such as *"a Community-supported record may not be used for FHIR exchange; it needs Expert-verified or higher"*. Because the gate reads the published file, editing the file changes the gate, and a test proves it.

3.3.5 FHIR packaging and profiles

`fhir.py` builds a collection Bundle for one verified report, and refuses unless the gate allows FHIR exchange. The Bundle contains a Location (position coarsened to about 100 m with a site identifier made of stream and 100 m cell), a Device (the evidence engine), an Organization, a Practitioner (the expert, by pseudonym), one Observation per sign and three Provenance resources (author, assembler, verifier with the organisation's signature). The Observation uses the OAH indicator as `code`, the citizen sign as `value`, and components for evidence grade, evidence score, trust level, independent corroborations and each permitted use. There is no Patient resource and no personal data.

The profile `StreamProofObservation` contains the invariant `sp-obs-1`, which makes the profile itself reject any record whose trust level is not Expert-verified or Decision-grade, even if an application skipped its own gate. An "everything looks fine" report is recorded with the SNOMED CT value *Absent*, and the DipteraCAST form states that a visual check supports "not seen", not "absent".

3.3.6 Signing and certificates

`signing.py` builds a de-identified view of the verified record (pseudonym, signs, area at three decimals, dates, grade, trust level, verifier), canonicalises it as sorted JSON, hashes it with SHA-256 and signs the hash with an Ed25519 key [8]. The contribution certificate (PDF) carries the hash and

signature; the public verification page recomputes the hash, so changing any value is reported as a hash mismatch. Because the hash never covered the citizen's identity, deleting the identity link later does not invalidate the evidence. This is a signature, not a blockchain, because a chain would make erasure harder rather than easier.

3.3.7 River Health Brief and coverage

`brief.py` builds the city-facing summary and sends every section through the gate: the signal table may show Assessed records (labelled unverified), the advisory section only Decision-grade ones. Each signal lists its confidence (Low, Building, Moderate or High), its place name along the stream and next steps from `catalogue.py`, which quotes the OAH Catalogue with section, title and page and states where the Catalogue is silent (it has no measure for foam or odour, so those entries say they are the team's inference). `coverage.py` splits each mapped stream into 500 m reaches and counts, per reach, how many *different people* reported in the last 30 days; a reach with no reports or only one person is listed as under-observed, and an organisation can send a community mission there.

3.3.8 DipteraCAST ground-truth export

`dipteracast.py` exports expert-verified records that say something about Diptera: a mosquito report is labelled *present* and a verified all-clear is labelled *not seen*. Records the gate blocks, or that say nothing about Diptera, are counted and skipped. The export is available as CSV or as an OAH-profiled FHIR Bundle, contains no observer and no exact GPS, and the prediction slot is an empty interface because no public DipteraCAST API exists yet.

3.3.9 Privacy and security

Citizens are pseudonyms with no password; exact position and contact details sit in an identity vault table, public surfaces and every FHIR Location use a position coarsened to about 100 m, stored photographs lose their EXIF data, and the "forget me" function deletes the vault entry while keeping the signed evidence valid. Reviewer accounts use scrypt-hashed passwords, failed sign-ins are rate limited, organisation reviewers see only their own city, and demo sign-in can be switched off with one environment variable. A real deployment would still need a data-protection officer's review under the GDPR [10].

3.3.10 Web application

The Next.js application provides the citizen screens (start, report, my reports, report card, share card, verification page, profile) and the organisation screens (review queue and detail, River Health Brief, dashboard, missions, team, Standards). The Standards page publishes the rule files, the permitted-

use matrix, the OAH mapping, the validation outcome including the negative control, and a calibration table computed live on whatever expert decisions exist.

3.4 Summary

This chapter described the data, the layered architecture, the trust graph and the permitted-use matrix, and the modules that implement grading, corroboration, gating, FHIR packaging, signing, the city brief and the DipteraCAST export. The central design decision is that the rule governing use is a published FHIR file that the running system loads, not a private piece of code.

CHAPTER 4
PERFORMANCE ANALYSIS AND RESULTS

This chapter separates three kinds of statement: what we measured ourselves while preparing this report, what the repository's own saved outputs record, and what has not been measured at all. All figures in Section 4.1 were produced by running the repository's test suite and an in-process timing script on the seeded data in a cloud Linux sandbox, so absolute times will differ on other machines.

4.1 Functional Performance

Test suite. The backend has 93 tests across seven files, and running them gave 88 passed and 5 skipped with no failures. Table 4.1 shows what each file covers.

Table 4.1: Backend test suite (93 tests; 88 passed, 5 skipped in our run)

Test file	Tests	What it checks
test_engine.py	24	Geometry, grading (caps, photo warnings, nearby agreement), corroboration rules and caps, expert shortcut, decision-grade thresholds, the gate, FHIR shape and privacy, signatures
test_oah.py	18	The gate is the published file; editing it changes the gate; CodeSystems match the code; ConceptMap matches the mapping; mapped codes exist in the official OAH system; DipteraCAST export rules
test_api.py	15	Full citizen-to-expert story, tamper detection, citizen sees only own reports, EXIF stripping, bad input, forget-me, missions
test_auth.py	15	Password hashing, roles, login rate limit, team admin, consent, share card, sign-up and sign-out
test_equity.py	9	Coverage reaches, under-observed counting by people, citizen updates, calibration table
test_catalogue.py	6	Every Catalogue title and quoted phrase checked against the Catalogue text
test_cities.py	6	City search, registering a city, grading on a newly fetched stream, scoping

The five skipped tests were reported by pytest as skipped; we did not investigate why, and we have not claimed that they pass.

Demo story. The end-to-end test `test_full_story` reproduces the intended use. A citizen submits a stagnant-water and mosquito report with a photograph and receives a grade of A or B with seven reasons and an inexact position; a reviewer sees the exact position and the gate; the FHIR export is refused with 403 and the message *"Blocked by the permitted-use gate"*; after an expert verifies, the export succeeds, a signed certificate (PDF) is produced, and editing a value on the verification page is detected as tampering.

Response time. We timed the API in-process (no network) on the 15 seeded reports with a 1200 × 900 pixel photograph. The figures are medians of repeated calls.

Table 4.2: Measured API timings (in-process, seeded data)

Operation	Median time	Remarks
Submit report (photo analysis + grading + corroboration)	about 135 ms	8 submissions, maximum about 176 ms
River Health Brief	about 119 ms	20 calls, maximum about 181 ms
Review queue	about 117 ms	20 calls
Standards page data	about 46 ms	20 calls
FHIR Bundle export	about 12 ms	14.5 KB; Location, Device, Organization, Practitioner, Observation, 3 Provenance
Certificate PDF	about 7 ms	10 calls
Verification check	about 4 ms	20 calls

These timings show that the rule engine is not a bottleneck at demonstration scale. They do not show how the system behaves under concurrent load or with a large database, which was not tested.

4.2 Accuracy and Reliability

Standards conformance (recorded in the repository). The repository stores the output of an HL7 Validator run (version 6.10.4, FHIR 4.0.1, against the official OAH package `hl7.eu.fhir.oah#0.1.0-ci-build`). We inspected the saved outcome file: it covers 18 resources (4 example Bundles and 14 definitions) and contains only informational messages, that is, 0 errors and 0 warnings. A fifth Bundle, the negative control (an unverified record), produces exactly one error, the invariant `sp-obs-1`. This shows that the profile itself rejects an unverified record. We did not re-run the validator, which needs Java and network access to a terminology server.

Rule behaviour (specification check). The repository's calibration report grades reports built to isolate one weakness. Figure 4.1 reproduces its scenarios. This shows that the rules behave as designed; it is *not* evidence that the grade predicts anything about the stream.

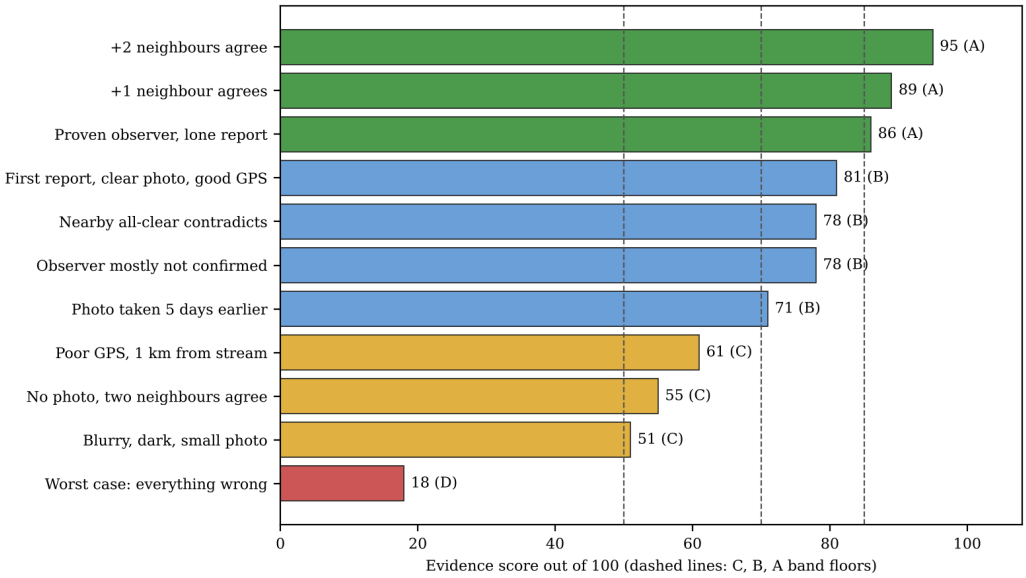


Figure 4.1: Score and grade of the specification scenarios (from the repository's calibration report)

Three behaviours deserve attention. A lone newcomer with a clear photograph and good GPS tops out at B (81/100), so grade A needs a second person or a proven track record. A single agreeing neighbour lifts that report from B to A (89/100), which is generous because agreement is cheap to fake; this is why the *trust level*, which counts different people spread in time or place and needs an expert for decision grade, is what the gate uses rather than the grade. A contradicting all-clear costs a report only 3 points, so contradictions are settled by the expert, not by the grade.

Does the grade predict expert decisions? This is the calibration that matters, and it cannot be answered yet. On the synthetic seeded data, 12 reports have been decided (1 grade A, 11 grade B; 11 confirmed), but the repository requires at least 30 decided reports before any rate is shown, and synthetic decisions would not measure real accuracy anyway. No accuracy, precision or recall is therefore claimed. The repository describes a pilot protocol in which ecologists decide a sample of reports of every grade, so that the sample is not biased towards health-relevant reports.

Resistance to gaming. Tests confirm that one observer cannot corroborate themselves, that two reports at the same spot in the same minute are not independent, that the daily corroboration cap applies, and that the decision threshold counts people, not reports. The honest limit is that accounts in the demonstration have no identity verification, so a determined person with many accounts could still reach Community-supported; they could not reach decision grade, or pass the FHIR gate, without an expert.

4.3 User Interaction and Experience

The citizen flow was designed to take about one minute: camera inside the page, one tap per sign, and the location snapping to the stream. Every grade is explained in plain words, an everything-looks-fine report is credited equally to a problem report so that people are not nudged towards reporting problems, and a report that is not confirmed carries no penalty. The application works offline through the service worker and the outbox, is translated into 45 languages, and the repository includes scripted axe-core accessibility audits for every page in light and dark mode on phone and desktop. We did not run the browser scripts for this report, so these properties are as documented by the repository, not independently confirmed here. No usability study with real citizens has been carried out.

4.4 System Limitations Observed

- **Synthetic demonstration data.** Reports, observers and verifications are invented; only stream geometry and rainfall are real. Hence no accuracy figure exists.
- **Illustrative thresholds.** Weights, bands and decision thresholds are configurable defaults, not validated values, and need calibration with OneAquaHealth ecologists.
- **No connection to OneAquaHealth systems.** No public API for the Citizen Science App was found; placing StreamProof behind it is a proposed adoption path, not an agreement. The DipteraCAST export is real, but its input is an empty interface.
- **Demo-grade identity.** Citizen accounts are pseudonymous without identity verification.
- **Machine-assisted translations.** The 45 languages have not been reviewed by native speakers and a few late screens are English only.
- **Advisory, never diagnostic.** A decision-grade flag says conditions “may warrant inspection”; it is not a diagnosis, and the grade is not the ecological status of the stream.
- **Selection effects in calibration.** Experts review health-relevant reports first and read the grade’s reasons before deciding, so a naive confirmation rate would be biased.
- **Scale untested.** SQLite and in-process timings say nothing about load, and no GDPR assessment has been done.

4.5 System Scalability and Future Enhancements

The architecture scales by replacing components rather than redesigning them: SQLite can be replaced by a server database, the gate and the profiles are already independent of the application,

and any city can be added by search because the stream geometry is fetched from OpenStreetMap. The following enhancements follow from the limitations above.

- Run a pilot with OneAquaHealth ecologists, record real expert decisions across all grades, publish the calibration table with its sample size and tune the weights and bands.
- Offer the add-on (trust-level, permitted-use and the proposed indicators) to HL7 Europe as an extension of the OAH guide, with the application as a reference client.
- Connect the DipteraCAST prediction slot when a public model API exists, and connect to the OAH Citizen Science App.
- Strengthen identity (for example verified organisation accounts) and complete a data-protection assessment.
- Add native-speaker review of translations and a usability study with citizens, and consider volunteer biotic indices as a further observation type.

4.6 Overall Outcome

StreamProof meets its design objectives as a working prototype: reports are graded with readable reasons, strengthened by independent people or an expert, gated by a rule that is the published FHIR file, and exported as FHIR that conforms to the official OAH profiles and refuses unverified records. The tests pass, the response times are small, and the limits are stated. What the project does not yet show is that its grade agrees with real expert judgement; that remains the first thing to measure in a pilot.

CHAPTER 5
SAMPLE CODE AND SCREENSHOTS

This chapter lists short extracts of the source code that carry the core ideas, followed by screenshots of the working application. The extracts are reproduced from the repository [13] with long lines wrapped and some comments shortened.

Listing 5.1: Evidence grading (app/grading.py)

```
GRADE_BANDS = [(85, "A"), (70, "B"), (50, "C"), (0, "D")]

def grade(r: Report, others: list[Report]) -> Report:
    """Fill grade, score, reasons, hint and safety on `r`."""
    reasons = [_photo(r), _photo_time(r), _location(r), _stream(r),
               _nearby(r, others), _context(r), _track(r, others)]
    score = sum(x.points for x in reasons)
    g = _band(score)
    if not r.photo and g in ("A", "B"):
        g = "C"                      # no photo: capped at C
    r.reasons, r.score, r.grade = reasons, score, g
    r.hint = _hint(reasons)
    r.safety = SAFETY_TEXT if set(r.indicators) & SAFETY_SIGNS else None
    if r.rung == Rung.REPORT:
        r.rung = Rung.ASSESSED
    return r
```

Listing 5.2: Independent corroboration (app/evidence.py)

```
def _independent(a: Report, b: Report) -> bool:
    """Different people, and spread in time or place."""
    if a.observer == b.observer:
        return False
    gap_min = abs((a.created_at - b.created_at).total_seconds()) / 60
    gap_m = geo.distance_m(a.lat, a.lon, b.lat, b.lon)
    return (gap_min >= config.MIN_CORROBORATION_GAP_MIN
            or gap_m >= config.MIN_CORROBORATION_GAP_M)

# inside link_corroboration(): weight each link by track record
old.supporters[new.id] = reliability(new.observer, others)
new.supporters[old.id] = reliability(old.observer, others)
if r.rung == Rung.ASSESSED and r.support >= COMMUNITY_THRESHOLD:
    r.rung = Rung.COMMUNITY
```

Listing 5.3: Permitted-use gate (app/permitted_use.py)

```
def check(rung: Rung, use: str) -> Decision:
    if use not in USES:
        raise KeyError(f"unknown use: {use}")
    if use in allowed_uses(rung):
        return Decision(True, use, rung,
                        f"{rung.label} may be used for: {USES[use]}.")
    if rung == Rung.NOT_CONFIRMED:
        return Decision(False, use, rung,
                        "Not-confirmed reports are kept for the record only.")
    need = minimum_rung(use)
    return Decision(False, use, rung,
        f"Blocked by the permitted-use gate: a {rung.label} record may not "
        f"be used for '{USES[use]}'. It needs {need.label} or higher.")

# the matrix itself is loaded from the published CodeSystem:
TRUST_LEVEL_FILE = config.RULES_DIR / "CodeSystem-trust-level.json"
```

Listing 5.4: FHIR profile with the trust invariant (ig/input/fsh/profiles.fsh)

```
Invariant: sp-obs-1
Description: "Only Expert-verified or Decision-grade records may be
              exchanged: nothing unverified can enter an OAH system."
Severity: #error
Expression: "component.where(code.coding.where(code = 'trust-level')
  .exists()).value.coding.where(code = 'expert-verified'
  or code = 'decision-grade').exists()"

Profile: StreamProofObservation
Parent: ObservationIndicatorsOah
* obeys sp-obs-1
* component contains evidenceGrade 1..1 and evidenceScore 1..1 and
    trustLevel 1..1 and independentCorroborations 1..1 and permittedUse 0..*
```

Listing 5.5: Tamper-evident certificate (app/signing.py)

```
def digest(d: dict) -> str:
    return hashlib.sha256(canonical(d)).hexdigest()

def sign(d: dict) -> dict:
    h = digest(d)
    sig = _load_or_create().sign(bytes.fromhex(h))   # Ed25519
    return {"sha256": h, "signature": base64.b64encode(sig).decode(),
            "alg": "Ed25519", "public_key": public_key_b64(),
            "signed_by": config.ORG_NAME}

def verify(d, sha256_hex, signature_b64, public_key=None):
    if digest(d) != sha256_hex:
        return False, "The record has changed since it was signed."
    ...
```

Listing 5.6: FHIR Bundle packaging (app/fhir.py)

```
def bundle(r: Report, signed=None, at=None, ground_truth=False) -> dict:
    """Refuses unless the gate allows FHIR exchange."""
    permitted_use.require(r.rung, "ground_truth" if ground_truth
                          else "fhir_exchange")
    lat, lon = geo.coarsen(r.lat, r.lon)      # about 100 m, never exact GPS
    ...
    components = [
        _component("evidence-grade", "Evidence grade", ...),
        _component("evidence-score", "Evidence score (0-100)", ...),
        _component("trust-level", "Trust level", ...),
        _component("independent-corroborations", ...),
    ] + [_component("permitted-use", "Permitted use", ...)
         for use in permitted_use.allowed_uses(r.rung)]
```

The screenshots below were taken from the running application and are taken from the repository's documentation. They show demonstration data.

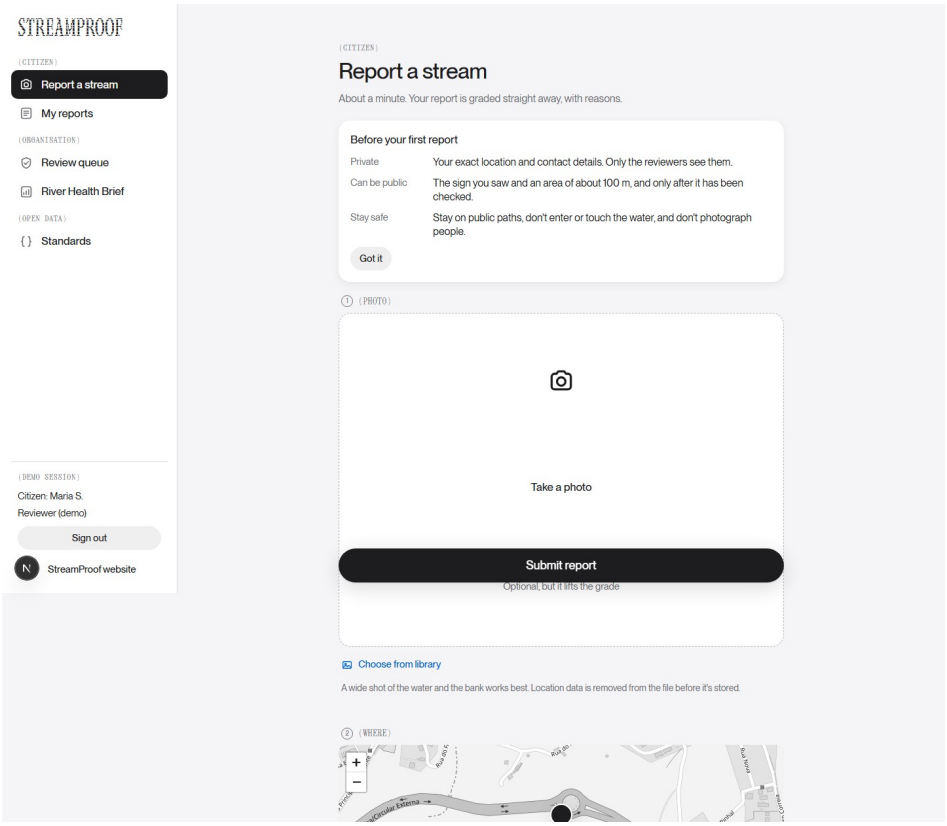


Figure 5.1: Report a stream: photograph, map, sign chips and notes (desktop)

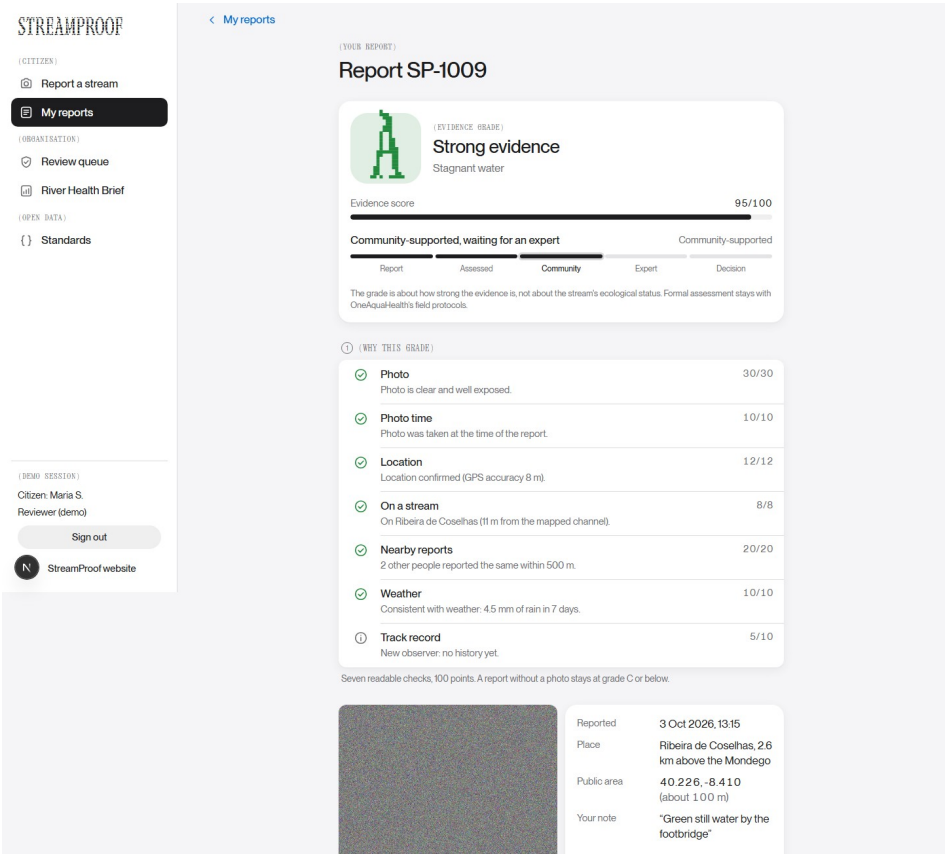


Figure 5.2: Report card: grade A with seven reasons, neighbours, and what the record may be used for

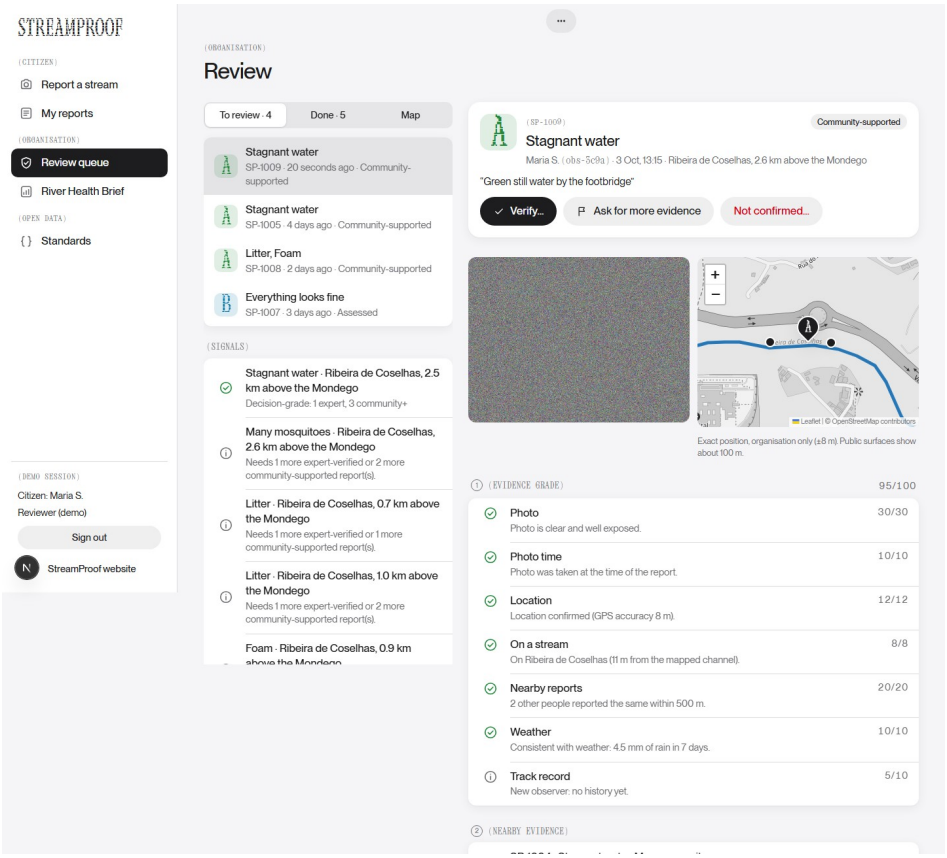


Figure 5.3: Reviewer screen: queue, map, evidence and the permitted-use gate with FHIR export blocked below Expert-verified

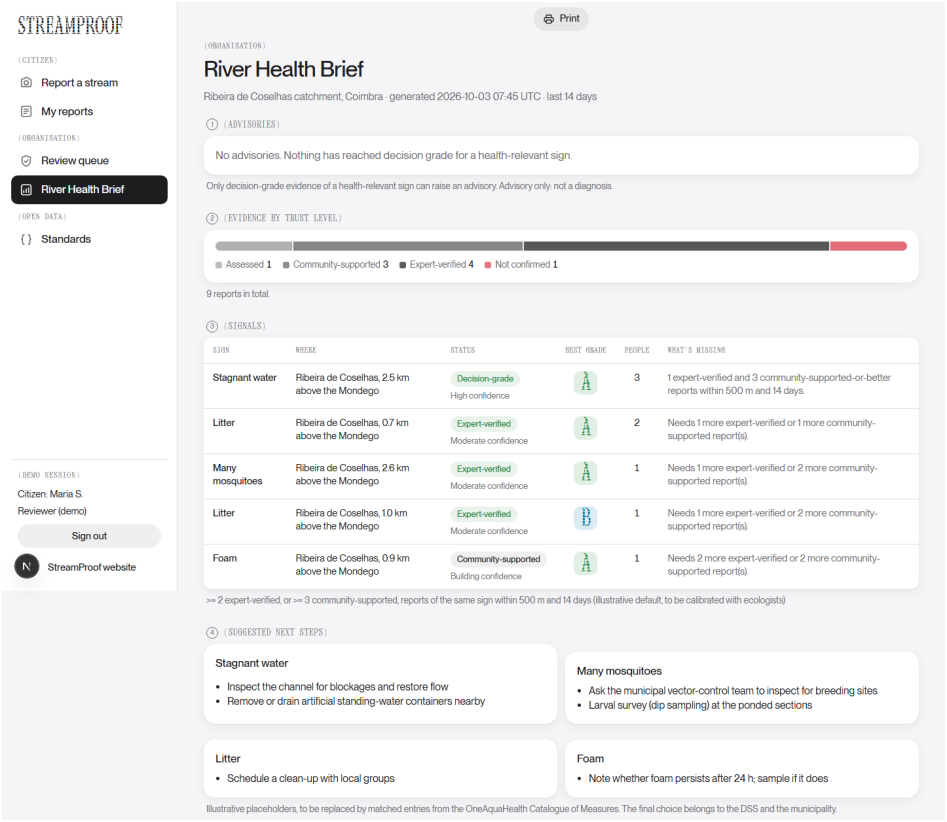


Figure 5.4: River Health Brief: signals, confidence, Catalogue-matched next steps and open missions

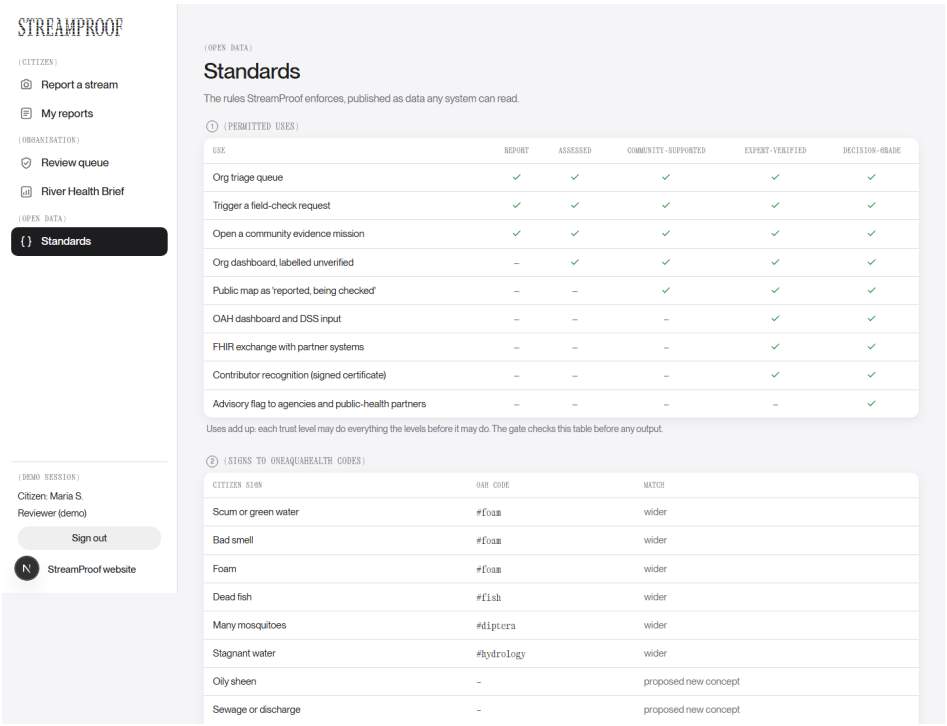


Figure 5.5: Standards page: published permitted-use matrix, OAH mapping and rule files

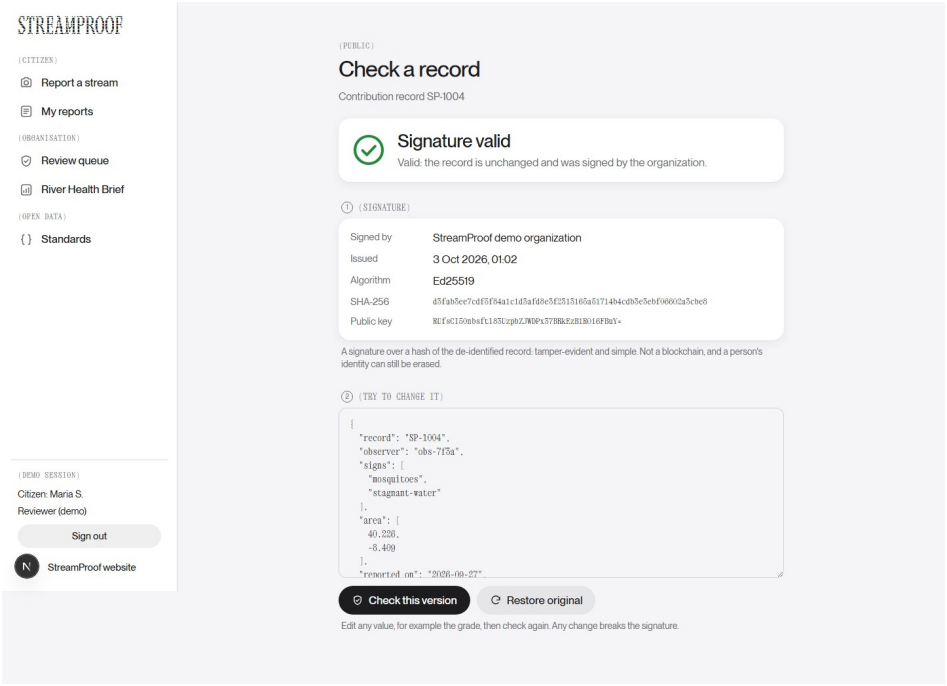


Figure 5.6: Certificate verification page: signature valid, with an option to tamper with a value

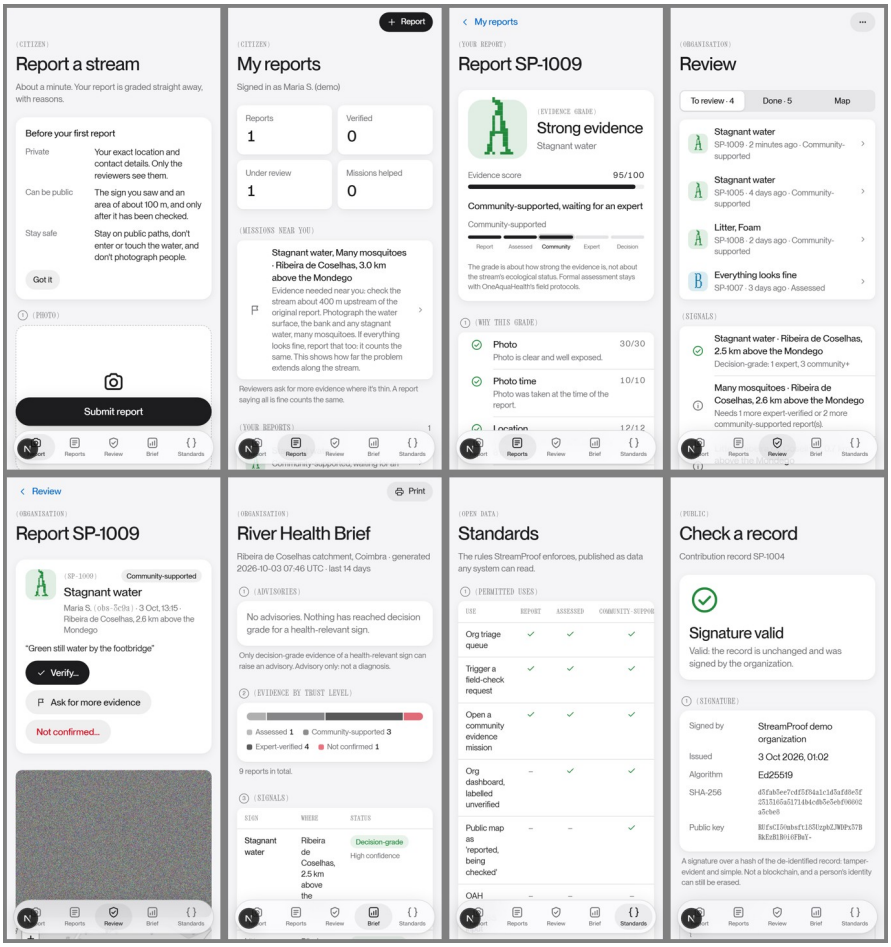


Figure 5.7: Phone layout of the citizen app (contact sheet of screens)

CHAPTER 6
CONCLUSION

Citizen reports about urban streams are plentiful but, as they stand, they are hard to trust, hard to exchange and silent about what they may be used for. StreamProof addresses this by grading each report with seven transparent checks, letting different people and experts strengthen it through an evidence graph, and publishing the use rule as a FHIR CodeSystem that the running system itself loads. Verified records leave as FHIR R4 Bundles whose profiles derive from the official OneAquaHealth profiles and carry grade, trust level and permitted uses inside the record, with a signed, de-identified certificate and privacy protections that survive erasure.

The results support the engineering claims: the gate refuses an export below Expert-verified and allows it afterwards, the saved validator run records 0 errors and 0 warnings for the add-on's Bundles and definitions, the unverified negative control is rejected by the profile, 88 of 93 backend tests passed in our run, and the main API operations took roughly 5 to 135 ms in-process. Equally important are the claims that are not made. The grade has not been compared with real expert decisions, thresholds are illustrative, the demonstration data are synthetic, no connection to OneAquaHealth systems exists yet, and no usability or load study has been done.

The next steps are therefore clear: a pilot with ecologists on real reports to calibrate the weights and measure agreement with expert decisions, an offer of the trust-level and permitted-use CodeSystems to HL7 Europe, integration with the OAH Citizen Science App and DipteraCAST once public interfaces exist, and a data-protection assessment. Until then StreamProof is best described as a working, tested and honestly bounded prototype of a trust layer for citizen-generated water evidence.

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